CO4-AT3 Open Data Analysis Project

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Global COVID-19 Trends and Vaccination Analysis

Aim :

To analyze global COVID-19 cases, deaths, and vaccination trends using Pandas, grouping, correlation analysis, dataset joining, and visualization.

Tools Used :

Python, Pandas, NumPy, Matplotlib, Seaborn

Procedure :

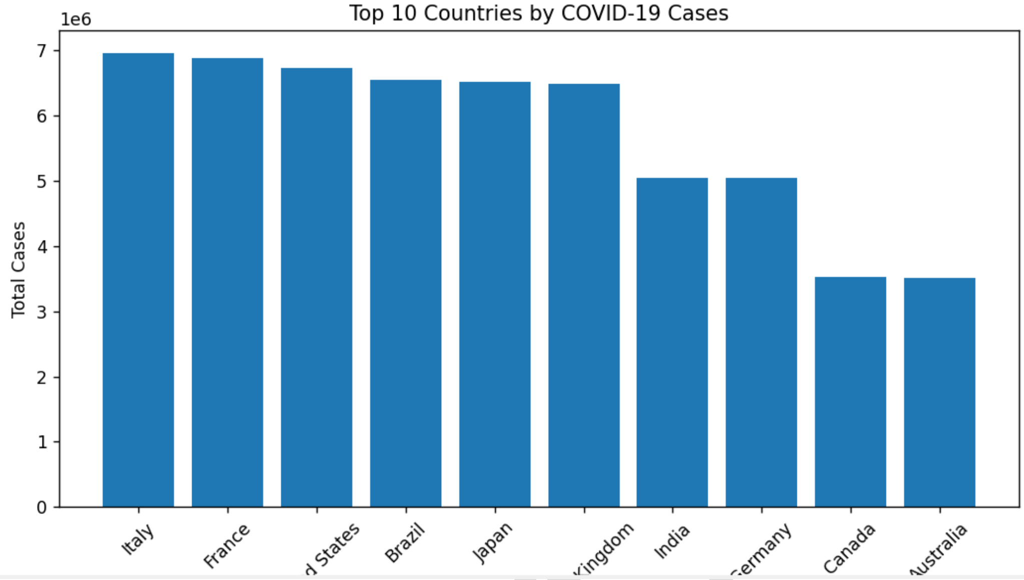
1. Load the COVID-19 dataset using Pandas.
2. Check missing values, duplicates, column types, and dates.
3. Clean and standardize the required columns.
4. Group data by country to compare cases, deaths, and vaccination.
5. Join COVID case/death data with vaccination data using location and date.
6. Calculate correlations between vaccination coverage and mortality.
7. Visualize trends using line charts, bar charts, and heatmaps.

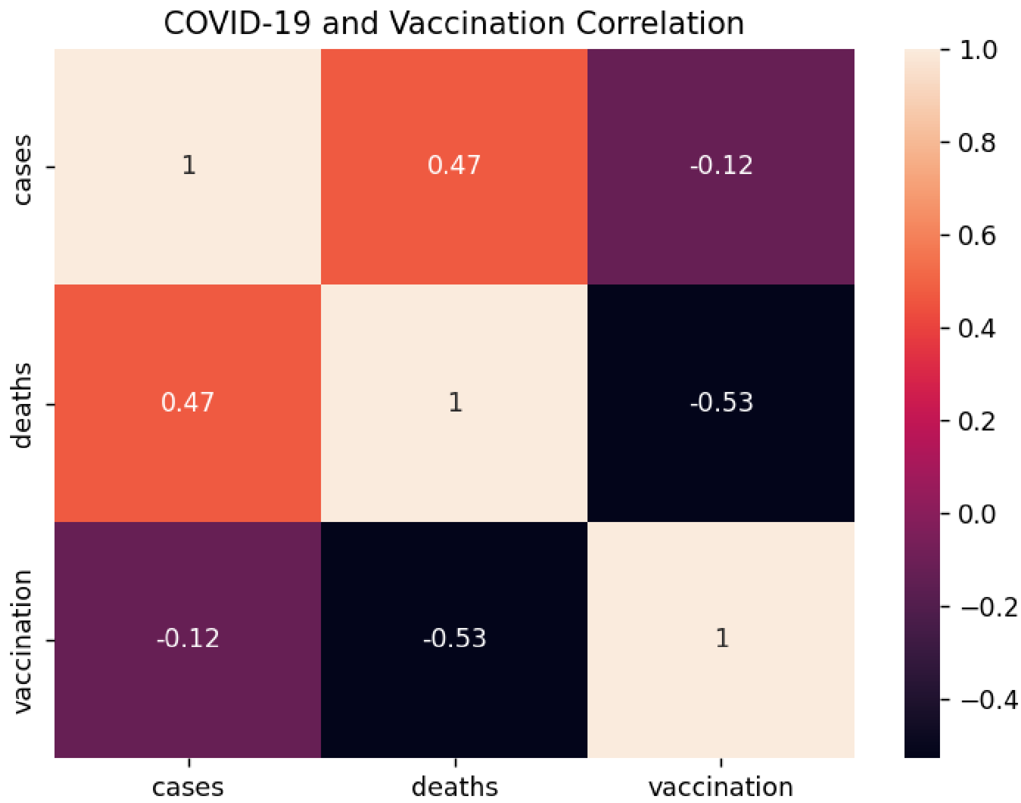
Python Code

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns


# Load dataset

df = pd.read_csv("owid-covid-data.csv")


# Select required columns

df = df[[

    "location", "date", "population",

    "total_cases", "total_deaths",

    "people_vaccinated", "people_fully_vaccinated"

]]


# Convert date

df["date"] = pd.to_datetime(df["date"])


# Check missing values and duplicates

print(df.isnull().sum())

print("Duplicates:", df.duplicated().sum())


# Calculate vaccination percentage

df["vaccination_rate"] = (

    df["people_vaccinated"] / df["population"]

) * 100


# Country-wise analysis

country = df.groupby("location").agg(
```

```
    cases=("total_cases", "max"),
    deaths=("total_deaths", "max"),
    vaccination=("vaccination_rate", "max")
).reset_index()

print(country.head())

# Correlation
print(country[[
    "cases", "deaths", "vaccination"
]].corr())

# Top 10 countries by cases
top = country.nlargest(10, "cases")

plt.figure(figsize=(10, 5))
plt.bar(top["location"], top["cases"])
plt.xticks(rotation=45)
plt.title("Top 10 Countries by COVID-19 Cases")
plt.ylabel("Total Cases")
plt.show()

# Correlation heatmap
plt.figure(figsize=(7, 5))
sns.heatmap(
    country[["cases", "deaths", "vaccination"]].corr(),
    annot=True
)
plt.title("COVID-19 and Vaccination Correlation")
plt.show()
```

OUTPUT :


Top 10 Countries by COVID-19 Cases
1e6
Total Cases
0
1
2
3
4
5
6
7
Italy
France
Brazil
Japan
India
Canada


COVID-19 and Vaccination Correlation
cases
deaths
vaccination
1
0.47
-0.12
0.47
1
-0.53
-0.12
-0.53
1
cases
deaths
vaccination
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

```
Duplicates: 0
    location    cases  deaths  vaccination
0  Australia  3518042   91757    69.372150
1     Brazil  6548164   89962    73.234737
2     Canada  3528222  125340    57.053868
3     France  6890237  192935    63.401721
4    Germany  5041881   90340    73.136189
                cases    deaths  vaccination
cases        1.000000  0.470264    -0.124903
deaths       0.470264  1.000000    -0.529483
vaccination -0.124903 -0.529483     1.000000
PS E:\Query process>
```

Key Insights

COVID-19 cases and deaths varied considerably between countries. Vaccination coverage differed significantly across regions. Grouping by country helps identify countries with higher case and mortality burdens. Correlation analysis helps identify the relationship between vaccination coverage and COVID-19 outcomes. Population-adjusted measures should be preferred when comparing countries of different sizes.

Conclusion

The analysis shows that Pandas and visualization techniques can effectively identify global COVID-19 trends and vaccination patterns. Grouping, joining, and correlation analysis provide useful evidence for comparing countries and understanding relationships between vaccination and pandemic outcomes. Correlation should be interpreted as an association and not as proof of causation.